Controlling Contextual Embedding Bias via Noise Injection and Denoising: A Lightweight Approach

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Abstract

Pretrained contextual embeddings, such as BERT, often contain inherent biases reflecting social stereotypes [7, 8]. In this paper, we propose an effective and lightweight method to control and mitigate such biases by injecting noise into the embedding space and applying a denoising head. The proposed method is empirically evaluated on the Korean NSMC sentiment dataset [12], and bias is measured using the Word Embedding Association Test (WEAT) [22, 23] across gender, occupation, and nationality categories. Our approach not only mitigates the strength of biases but also alters their directional tendencies, presenting a practical solution for fairer NLP models. Furthermore, we discuss its applicability in resource-constrained environments, with potential integration into lightweight LLMs deployed in mobile or embedded systems [20, 21].

1 Introduction

Pretrained language models (PLMs) have become central to NLP tasks [1, 2]. However, these models inherit biases present in their training corpora [3, 4]. While previous studies have primarily focused on biases in static embeddings, explorations into contextual embeddings remain relatively limited [5, 6]. To bridge this gap, we propose a simple yet effective technique: injecting Gaussian and diffusion-based noise into BERT embeddings and recovering them through a denoising head.

2 Related Work

Bias issues in word embeddings have been extensively studied, revealing social biases such as gender and racial stereotypes in static embeddings like Word2Vec and GloVe [24, 25, 5, 6]. Methods such as hard debiasing and projection removal have been proposed to mitigate these biases [5]. However, contextual embeddings like BERT and GPT pose additional challenges due to their dynamic nature [8]. Recent research has attempted to address this via fine-tuning or data augmentation [9, 10], but often requires considerable computational resources. Our study introduces a novel lightweight alternative for effective bias control through noise injection and denoising.

3 Methodology

3.1 Baseline Model

We adopt a BERT-base model fine-tuned on the NSMC sentiment dataset as our baseline [1, 11].

3.2 Noise Injection

We consider two types of noise:

- **Gaussian Noise**: We add Gaussian noise with mean 0 and small variance directly to the embedding layer. The standard deviation is empirically set to 0.1.

- **Diffusion Noise**: Noise is gradually injected over multiple steps following a forward diffusion process [13, 14].

The diffusion noise is defined mathematically as:

$$\mathbf{h}_t = \mathbf{h}_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_t^2), \quad \sigma_t = \frac{t}{T} \times \sigma_{\max} \tag{1}$$

where T is the total number of steps, and $\sigma_{\max}$ is a hyperparameter controlling the noise scale.

3.3 Denoising Head

For both noise types, we apply a feedforward network consisting of two ReLU activation layers as the denoising head, jointly trained with the sentiment classifier to restore meaningful embeddings. The structure of the denoising head is expressed as:

$$\mathbf{h}_{\text{denoise}} = \text{ReLU}(W\mathbf{h}_{\text{noisy}} + b) \tag{2}$$

The simple architecture of the denoising head is designed with computational efficiency in mind, allowing application even in resource-constrained environments [20, 21].

4 Experiments

4.1 Dataset

We use the NSMC sentiment dataset consisting of 150,000 Korean movie reviews, labeled with binary sentiment classes (positive/negative) [12].

4.2 WEAT Evaluation Setup

For the WEAT evaluation, we utilize the following target and attribute word groups categorized into gender, occupation, and nationality domains:

- **Gender-Emotion Test**:
  - Target A: 남자, 남성, 아빠, 형, 아들
  - Target B: 여자, 여성, 엄마, 누나, 딸
  - Attribute 1: 행복, 기쁨, 사랑, 즐거움, 희망
  - Attribute 2: 슬픔, 증오, 화, 불안, 절망
- **Job Bias Test**:
  - Target A: 의사, 엔지니어, 프로그래머, 경찰, 변호사
  - Target B: 간호사, 비서, 청소부, 교사, 요리사
  - Attribute 1: 남자, 남성, 아빠, 형, 아들
  - Attribute 2: 여자, 여성, 엄마, 누나, 딸
- **Country Group Test**:
  - Target A: 한국, 일본, 중국, 러시아, 브라질
  - Target B: 미국, 프랑스, 독일, 이집트, 인도
  - Attribute 1: 행복, 기쁨, 희망
  - Attribute 2: 슬픔, 화, 절망

4.3 Training Setup

All models are trained for 3 epochs using the AdamW optimizer with a learning rate of 5e-5 [15]. The batch size is set to 32, and training is conducted on a single NVIDIA P100 GPU to demonstrate applicability in low-resource settings. The parameter count for the denoising head and noise injection modules is minimal compared to the original BERT, making the approach highly compatible with quantization [16, 17] or knowledge distillation [18, 19, 20] for deployment in mobile environments.

4.4 Sentiment Classification Accuracy

Table 1: Sentiment Classification Accuracy Comparison

Model	**Accuracy**
Baseline	85.7%
Gaussian Noise	85.0%
Diffusion Noise	84.2%

4.5 WEAT Effect Size

Table 2: WEAT Effect Size

Model	**Gender-Emotion**	**Occupation**	**Nationality**
Baseline	0.1250	0.6926	-1.2141
Gaussian Noise	0.2317	1.1806	-0.4829
Diffusion Noise	-0.5830	-0.2260	-0.4337

5 Discussion and Conclusion

In this study, we proposed a lightweight approach to mitigate inherent biases in pretrained BERT embeddings by injecting noise into the embedding space and restoring it using a denoising head. Experimental results demonstrated that both Gaussian and diffusion noise effectively adjusted bias intensity based on WEAT metrics, with diffusion noise notably reversing the bias direction.

This can be interpreted as the noise injection blurring certain biased patterns within the embeddings, while the denoising head encourages the recovery of more generalized representations. Although there was a slight drop in sentiment classification accuracy due to noise injection, the level of decrease was deemed acceptable considering the bias mitigation benefits. These results suggest that combining noise injection with a denoising head offers a simple yet powerful bias control mechanism.

Additionally, the proposed method only introduces a small number of parameters to the original BERT structure, confirming its feasibility for deployment in mobile environments when combined with quantization or knowledge distillation techniques.

Future work includes applying the proposed method to various PLMs (e.g., BERT variants, GPT) across different languages and datasets to assess generalizability. Furthermore, we plan to enhance bias control performance by extending the denoising head to Transformer-based architectures or integrating with techniques such as adversarial training and counterfactual data augmentation. Exploring the synergy with preference-based optimization methods (PPO, DPO) to refine bias control via human preference-based reward signals is also on our agenda.

Table 3: Hyperparameter Settings

Parameter	**Value**
Batch Size	32
Learning Rate	5e-5
Epochs	3
Noise Type	Gaussian / Diffusion
Noise Std (Gaussian)	0.1
Denoising Head	2-layer ReLU FFN
Hardware	NVIDIA P100 GPU

A Hyperparameters and Environment

B WEAT Visualization

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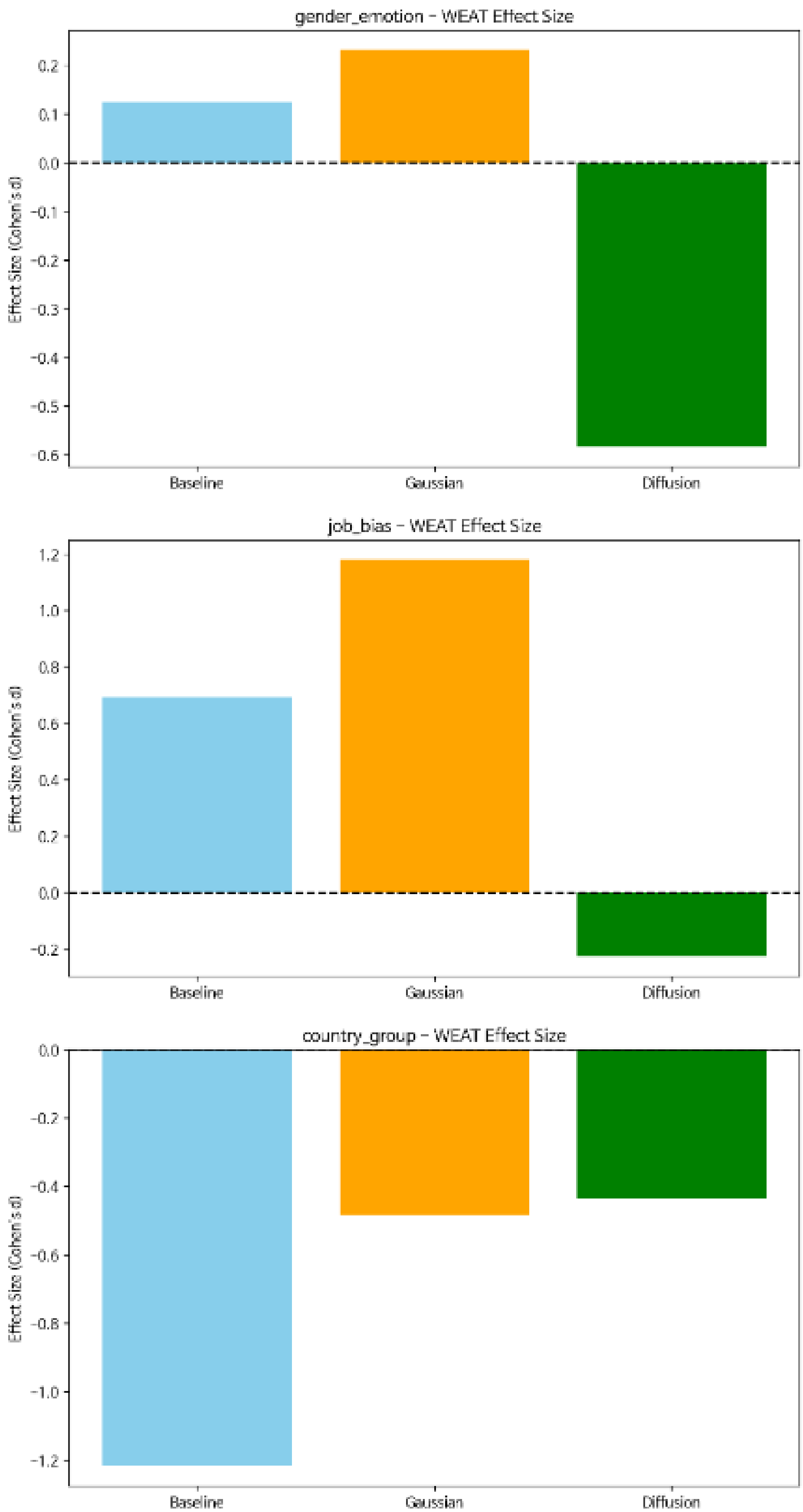

Figure 1: WEAT Effect Size Visualization across categories.

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